Types of variable

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Variables

- Characteristics that varies; Not constant
- **Values** Can be text or number
- Mostly used codes for the text values
- Variables in columns
- Observations in row
- Values of each observation for each variable in cell

First seven lines of data from a study of outcome after diagnosis of tuberculosis

ID	Hospital	Date of birth	Gender	Date of diagnosis	Weight(kg)	Smear result	Culture result	Skin test diameter (mm)	Alive after 6 months?	Days of hospital stayed	Completed age at diagnosis	Age group
001	A	1929-12-03	M	1998-08-23	56.3	Positive	Negative	18	Y	3	68	60 - 70
002	A	1936-04-13	M	1998-09-12	73.5	Positive	Negative	15	Y	4	62	60 - 70
003	A	1931-10-31	F	1999-06-17	57.6	Positive	Positive	21	N	5	67	60 - 70
004	B	1922-11-11	F	1999-07-05	65.6	Uncertain	Positive	28	Y	5	76	70 and elder
005	B	1946-05-01	M	1999-08-20	81.1	Negative	Positive	6	Y	6	53	50 - 60
006	C	1954-02-18	M	1999-09-17	56.8	Positive	Negative	12	Y	7	45	Younger than 50
007	33	1829-12-03	X	1998-08-23	99.0			NA	YY	NA	168	70 and elder

Types of Variable

- **Categorical variables**
  - Binary variable
  - Nominal variable
  - Ordinal variable
- **Numerical variables**
  - Discrete variable
  - Continuous variable

Binary variable

ID	Culture results	Alive after 6 months
001	Negative	Y
002	Negative	Y
003	Positive	N
004	Positive	Y
005	Positive	Y
006	Negative	Y
007		YY

Nominal variable

ID	Hospital name	Gender
001	A	M
002	A	M
003	A	F
004	B	F
005	B	M
006	C	M
007	33	X

Ordinal variable

ID	Smear result	Age group
001	Positive	60 - 70
002	Positive	60 - 70
003	Positive	60 - 70
004	Uncertain	70 and elder
005	Negative	50 - 60
006	Positive	Younger than 50
007		70 and elder

Discrete variable

ID	Days hospitalized
001	3
002	4
003	5
004	5
005	6
006	7
007	NA

Continuous variable

ID	Weight in kilogram	Skin test diameter(mm)	Age at diagnosis(years)
001	56.3	18	68
002	73.5	15	62
003	57.6	21	67
004	65.6	28	76
005	81.1	6	53
006	56.8	12	45
007	99.0	NA	168

Classifcation of variable

- can be **numerical** or **categorical**
- includes:
  - Recorded variables
  - Rates
  - Derived variable
  - Transformed variable

Rates

- occurrence over time
- usually follow exponential distribution
- not cover in my topics

Derived variable

- calculated or categorized from recorded variable
- categorized based on threshold value
- derived from reference curve/ standard values in population

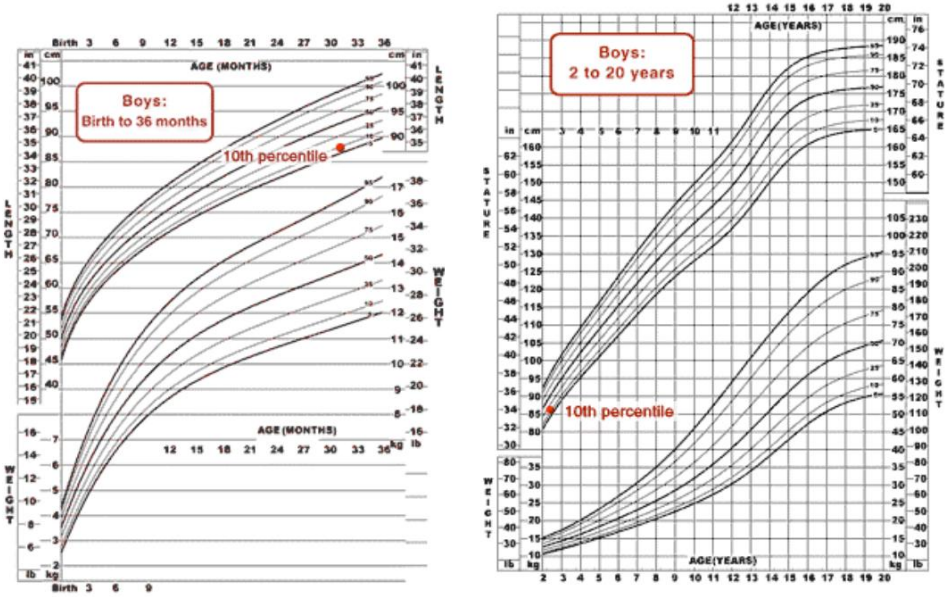
Calculated or categorized from recorded variable

id	date_of_birth	date_of_diagnosis	age_at_diagnosis_year	age_group
001	1929-12-03	1998-08-23	68	60 - 70
002	1936-04-13	1998-09-12	62	60 - 70
003	1931-10-31	1999-06-17	67	60 - 70
004	1922-11-11	1999-07-05	76	70 and elder
005	1946-05-01	1999-08-20	53	50 - 60
006	1954-02-18	1999-09-17	45	Younger than 50
007	1829-12-03	1998-08-23	168	70 and elder

Categorized based on threshold values

Derived variables	Original variable
Low birth-weight	Birth-weight
Yes	<2500g
No	>=2500g
Vitamin A status	Serum retinol level
Severe deficiency	<0.35umol/l
Mild/moderate deficiency	0.35-0.69umol/l
Normal	>=0.70umol/l

Derived from reference curve



Growth charts for boys(Image resource : CDC)

Transformed variable

- Variables with observations that violates assumptions
  - Normality assumption
  - Linear assumption
  - Homogeneity assumption
- Types of transformation
  - Logarithmic ($\log_e x$)
  - Reciprocal ($\frac{1}{x}$)
  - Square root ($\sqrt{x}$)
  - Square/cubic (x^n)

Distinguishing Outcome and exposure variables

- To choose appropriate data display and statistical method
- Outcome : seek to understand variation or occurrence
- Exposure : influence size or occurrence of outcome
- Statistical analysis : quantify the magnitude of association between exposures and outcomes

Examples of Outcomes and Exposures

Outcome	Exposure
Baby born with low birth weight (yes, no)	Mother smoked during pregnancy (yes, no)
Anthropometric status at 1 year of age (weight-for-age z-score)	Duration of exclusive breastfeeding (weeks)
Number of diarrhoea episodes experienced in a year	Access to clean water supply (yes, no)
Child develops leukaemia (yes, no)	Proximity to nuclear power station (miles)
Survival time (months) following diagnosis of lung cancer	Socio-economic status(6 groups)

Thank you